

FDFD Method Based on Refined Shift-and-Invert Arnoldi Technique for Periodic Leaky-Wave Antennas

Feng Xu, *Senior Member, IEEE*, and Wei Hong, *Fellow, IEEE*

Abstract—The finite difference frequency domain (FDFD) method can be used to analyze the properties of arbitrary complex close/open periodic guided-wave structures. For the open problem, it probably results in a large scale nonsymmetrical generalized eigenvalue problem. In the analysis of the leaky-wave antennas, because the leakage constant (attenuation constant) increases a lot in the working frequency range, the extraction process of the eigenvalues (complex propagation constants) is difficult to converge when the shift-and-invert Arnoldi technique is used to resolve this large scale nonsymmetrical generalized eigenvalue problem. In this paper, a refined shift-and-invert Arnoldi algorithm is adopted to speed the convergence rate by using refined Ritz vectors to approximate the desired eigenvectors. Since the refined method always converges, a simple FDFD method with six field components can be used. The difference equations of this method are much simpler than those of the previous method because the longitudinal field components don't have to be eliminated in order to approximately transform the generalized eigenvalue problem to a standard eigenvalue problem. Consequently, the accuracy of the method can be improved and the boundary conditions can be implemented easily.

Index Terms—Finite difference frequency domain (FDTD) method, large nonsymmetrical generalized eigenvalue problem, leaky-wave antenna, periodic guided-wave structure, refined shift-and-invert Arnoldi technique.

I. INTRODUCTION

NOWADAYS, more and more attentions have been put to the periodic guided-wave structure usually in two directions. One is related to using its normal guided-wave characteristics which are similar to those of the uniform guided-wave structure in order to obtain some new deformed guided-wave structures for example, substrate integrated waveguides (SIW) [1], [2]. Another is on the subject of its singular characteristics to form some new structures or materials for example, artificial

magnetic conductors, metamaterials, periodic leaky-wave antennas and so on. As a result, it is very important to determine and analyze the dispersion characteristics of a periodic guided-wave structure in practical RF engineering design. In slow-wave structures, backward-wave oscillators, corrugated antennas and leaky-wave antennas, periodic structures have been measured and investigated by many researchers due to their importance [3]–[7]. Accurate prediction of the propagation behavior of periodic structures is an essential step in the successful design of these systems. A variety of numerical techniques have been used to analyze periodic guided-wave structures. Among them, the finite difference frequency domain (FDFD) method is one of the most versatile techniques for solving partial differential equations and it has been used to analyze modal characteristics of arbitrarily periodic guided-wave structures [8]–[10]. It is much easier to extract the guided-wave characteristics in the first case than the second because the singular characteristics should be extracted in the second case.

Basically, a leaky-wave antenna can be regarded as direct geometrical evolution from a guided-wave structure which permits to leak energy along its longitudinal direction. As a result, the FDFD method can also be used to analyze periodic leaky-wave antennas (in this case, attenuation constant is called as leakage constant). When the FDFD method is used to extract the complex propagation constants, it will result in a large scale nonsymmetrical generalized eigenvalue equation. In this case, usually a shift-and-invert Arnoldi technique is introduced [11]–[13] to extract the propagation constants of the generalized eigenvalue equation close to a given shift σ . However, in the working frequency band of leaky-wave antennas, the leakage constants should increase in order to radiate energy to outside. The searching range of eigenvalues has to be enlarged to obtain the eigenvalues of interest. Consequently, the convergent process is quite slow, sometimes, failure to compute all the desired eigenvalues. In order to correct this problem, a refined shift-and-invert Arnoldi technique [14]–[16] is introduced to this electromagnetic analysis in order to guarantee and speed the convergence of computing the desired propagation properties (eigenvalues and eigenvectors). The refined method uses refined Ritz vectors with minimal residuals to approximate the desired eigenvectors. The refined Ritz vectors can be obtained by some small-sized singular value decompositions.

When the FDFD method is used to calculate the complex propagation constants, there are two opposite processing techniques which can be used to extract the parameters of interest. One is related to using the frequency as the input parameter to

Manuscript received May 03, 2012; revised November 02, 2012; accepted December 26, 2012. Date of publication January 03, 2013; date of current version May 29, 2013. This work was supported in part by the National Natural Science Foundation of China under Grant 61171052 and the specially appointed professor program foundation of Jiangsu province of China.

F. Xu is with the School of Electronic Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing, 210003, China (e-mail: feng.xu@njupt.edu.cn).

W. Hong is with the State Key Lab. of Millimeter Waves, School of Information Science and Engineering, Southeast University, Nanjing, 210096, China (e-mail: weihong@seu.edu.cn).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TAP.2013.2237741

extract the complex propagation constant [8]. In order to simplify the simulation model, the longitudinal field components are eliminated and only four transverse field components are involved in the final difference equations. However, the finite difference equations become much more complex and the implementation of the boundary conditions is indirect. Moreover, the condition that the mesh number in propagation direction must be odd in order to transform the generalized eigenvalue problem to a standard eigenvalue problem has confined the versatility of the FDFD method. Another technique is concerning with calculating the eigen-frequencies of interest when the initial input parameter is the phase constant β [10]. This FDFD method is constructed on the basis of an equivalent resonant cavity model. Although the finite difference equations are very simple and a standard eigenvalue problem is introduced, a post-processing must be performed to obtain the attenuation constant because the calculation of attenuation constant is indirect [10]. By using the refined shift-and invert Arnoldi method, we can quickly obtain the eigenvalues close to a given shift and the associated eigenvectors of a large symmetric or nonsymmetrical matrix pair. As a result, the simple FDFD method with original six field components can be used in the extraction of the complex propagation constants of periodic guided-wave structures, even in the case of the periodic leaky-wave antennas. Because the longitudinal field components don't need to be eliminated, the final finite difference equations are much simpler than those of literatures [8], [9]. The calculation of complex propagation constants is direct and convenient when compared with the method of literature [10].

II. FORMULATIONS

Supposing that the propagation direction is in z -direction for a periodic guided-wave structure, according to the Floquet's theorem for periodic structures [3], the electromagnetic field components can be expressed as

$$\vec{E}(x, y, z, t) = \vec{e}(x, y, z) e^{-\gamma z} e^{j\omega t} \quad (1a)$$

$$\vec{H}(x, y, z, t) = \vec{h}(x, y, z) e^{-\gamma z} e^{j\omega t} \quad (1b)$$

where γ is the propagation constant ($\gamma = \alpha + j\beta$), $e(x, y, z)$ and $h(x, y, z)$ is a periodic function with respect to z . If the periodic length of a periodic guided-wave structure is d , then

$$\vec{e}(x, y, z + d) = \vec{e}(x, y, z) \quad (2a)$$

$$\vec{h}(x, y, z + d) = \vec{h}(x, y, z) \quad (2b)$$

If the frequency is input parameter for solving the complex propagation constant, by substituting (1) into the Maxwell's differential equations, one can get

$$\gamma e_y = -(j\omega\mu + \sigma_x^m) h_x + \frac{\partial e_y}{\partial z} - \frac{\partial e_z}{\partial y} \quad (3a)$$

$$\gamma e_x = (j\omega\mu + \sigma_y^m) h_y + \frac{\partial e_x}{\partial z} - \frac{\partial e_z}{\partial x} \quad (3b)$$

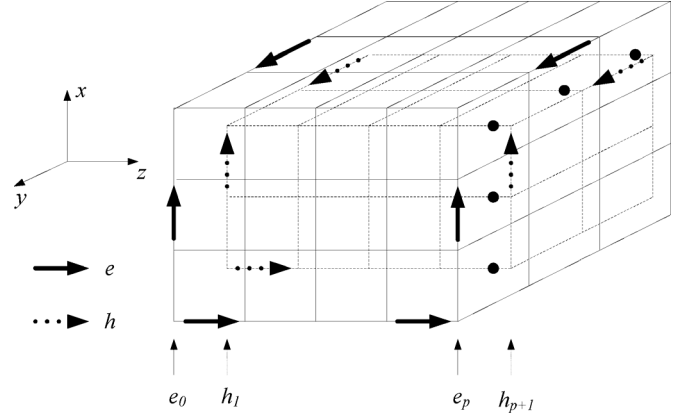


Fig. 1. Yee meshes of single period.

$$0 = -(j\omega\mu + \sigma_z^m) h_z + \frac{\partial e_x}{\partial y} - \frac{\partial e_y}{\partial x} \quad (3c)$$

$$\gamma h_y = (j\omega\epsilon_x + \sigma_x) e_x + \frac{\partial h_y}{\partial z} - \frac{\partial h_z}{\partial y} \quad (4a)$$

$$\gamma h_x = -(j\omega\epsilon_y + \sigma_y) e_y + \frac{\partial h_x}{\partial z} - \frac{\partial h_z}{\partial x} \quad (4b)$$

$$0 = (j\omega\epsilon_z + \sigma_z) e_z + \frac{\partial h_x}{\partial y} - \frac{\partial h_y}{\partial x} \quad (4c)$$

where σ and ϵ are the electric conductivity and permittivity of the medium, respectively. σ^m and μ are magnetic conductivity and permeability of the medium, respectively.

As shown in Fig. 1, the periodic boundary conditions are disposed along z direction. If the number of meshes in z direction is p and the space step in z direction is Δz , from (2), the periodic boundaries can be written as

$$e_0(x, y, 0) = e_p(x, y, d) \quad (5a)$$

$$h_{p+1}(x, y, d + \Delta z/2) = h_1(x, y, \Delta z/2). \quad (5b)$$

The difference equation of (3a) can be regarded as

$$\begin{aligned} & \frac{\gamma}{2} \left(e_y \left(l, m, n - \frac{1}{2} \right) + e_y \left(l, m, n + \frac{1}{2} \right) \right) \\ &= -(j\omega\mu + \sigma_x^m(l, m, n)) h_x(l, m, n) \\ &+ \frac{1}{\Delta z} \left(e_y \left(l, m, n + \frac{1}{2} \right) - e_y \left(l, m, n - \frac{1}{2} \right) \right) \\ &- \frac{1}{\Delta y} \left(e_z \left(l, m + \frac{1}{2}, n \right) - e_z \left(l, m - \frac{1}{2}, n \right) \right). \end{aligned} \quad (6)$$

Other finite difference equations are similar with (6). Obviously, the finite difference equations for all six field components are greatly simple and the boundary conditions can be easily applied. After that, an eigenvalue problem can be obtained

$$\gamma \mathbf{B} \mathbf{x} = \mathbf{A} \mathbf{x} \quad (7)$$

where $\mathbf{x} = [\mathbf{h}_x, \mathbf{h}_y, \mathbf{h}_z, \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z]^T$, if the electromagnetic waves propagate along z direction.

A. Structures of Matrices

From (3) and (7), the matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} \Delta & \Theta & \Theta & \Theta & \mathbf{A}_{15} & \mathbf{A}_{16} \\ \Theta & \Delta & \Theta & \mathbf{A}_{24} & \Theta & \mathbf{A}_{26} \\ \Theta & \Theta & \Delta & \mathbf{A}_{34} & \mathbf{A}_{35} & \Theta \\ \Theta & \mathbf{A}_{42} & \mathbf{A}_{43} & \Delta & \Theta & \Theta \\ \mathbf{A}_{51} & \Theta & \mathbf{A}_{53} & \Theta & \Delta & \Theta \\ \mathbf{A}_{61} & \mathbf{A}_{62} & \Theta & \Theta & \Theta & \Delta \end{bmatrix} \quad (8)$$

where Δ is a diagonal matrix, Θ is zero matrix, and each block matrix $\mathbf{A}_{mn}$ is bidiagonal matrix. Obviously, the matrix $\mathbf{A}$ is large sparse matrix. The vector $\mathbf{x} = [\mathbf{h}_x, \mathbf{h}_y, \mathbf{h}_z, \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z]^T$ is always disposed by first starting along the periodic direction (z direction). The matrix $\mathbf{B}$ is also a large sparse matrix

$$\mathbf{B} = \begin{bmatrix} \Theta & \Theta & \Theta & \Theta & \mathbf{B}^h & \Theta \\ \Theta & \Theta & \Theta & \mathbf{B}^h & \Theta & \Theta \\ \Theta & \Theta & \Theta & \Theta & \Theta & \Theta \\ \Theta & \mathbf{B}^e & \Theta & \Theta & \Theta & \Theta \\ \mathbf{B}^e & \Theta & \Theta & \Theta & \Theta & \Theta \\ \Theta & \Theta & \Theta & \Theta & \Theta & \Theta \end{bmatrix} \quad (9)$$

where $\mathbf{B}^e$ or $\mathbf{B}^h$ is also a bidiagonal matrix.

As a result, it can be concluded that the matrix $\mathbf{B}$ is a large nonsymmetrical sparse singular matrix. By means of the refined shift-and invert Arnoldi technique, the eigenvalues (complex propagation constants) of this generalized eigenvalue problem can be extracted accurately and quickly.

B. Arnoldi Method

The Arnoldi method [17] is an efficient procedure for approximating a subset of the eigensystem of a large sparse $n \times n$ matrix $\mathbf{C}$. The Arnoldi method is a generalization of the Lanczos process and reduces to that method when the matrix $\mathbf{C}$ is symmetric. After k steps, the method produces an upper Hessenberg matrix $\mathbf{H}_k$ of order k . The eigenvalues of this small matrix $\mathbf{H}_k$ are used to approximate a subset of the eigenvalues of the large matrix $\mathbf{C}$. The matrix $\mathbf{H}_k$ is an orthogonal projection of $\mathbf{C}$ onto a particular Krylov subspace, and the eigenvalues of $\mathbf{H}_k$ are usually called Ritz values or Ritz approximations. The method builds an orthogonal basis for the Krylov space

$$\mathbf{K}_k(\mathbf{C}, \mathbf{v}_1) \equiv \text{span}\{\mathbf{v}_1, \mathbf{C}\mathbf{v}_1, \dots, \mathbf{C}^{k-1}\mathbf{v}_1\} \quad (10)$$

for $\mathbf{C}$ generated by the vector $\mathbf{v}_1$ (usually chosen randomly).

After k steps, the Arnoldi method calculates a truncated factorization

$$\mathbf{C}\mathbf{V}_k = \mathbf{V}_k\mathbf{H}_k + h_{k+1,k}\mathbf{v}_{k+1}\mathbf{e}_k^* = \mathbf{V}_{k+1}\tilde{\mathbf{H}}_k \quad (11)$$

of $\mathbf{C} \in \mathbf{C}^{n \times n}$ into upper Hessenberg form where $\mathbf{V}_k^* \mathbf{V}_k = \mathbf{I}_k$. The vector $h_{k+1,k}\mathbf{v}_k$ is the residual and is orthogonal to the columns of $\mathbf{V}_k$. The matrix $\mathbf{H}_k \in \mathbf{C}^{k \times k}$ is an upper Hessenberg matrix that is the orthogonal projection of $\mathbf{C}$ and $\mathbf{V}_{k+1} = (\mathbf{V}_k, \mathbf{v}_{k+1})$. The eigenvalues (Ritz values) of the matrix $\mathbf{H}_k$ will be some eigenvalues of interest of the matrix $\mathbf{C}$ when these Ritz values converge.

C. Shift-and-Invert Arnoldi Method

In the application of periodic guided-wave structures, one is interested in only computing some eigenvalues in the complex plane, which are usually the complex propagation constants. One of the most commonly used techniques for this kind of large scale generalized nonsymmetrical eigenvalues problem is the shift-and-invert Arnoldi method [11]–[13]. Usually, the use of spectral transformations [18]–[21] is closely related to the inverse power and inverse iteration techniques. They are typically designed to enhance convergence to eigenvalues near a selected point in the complex plane. For (7), if some eigenvalues of interest are close to a shift σ , we have

$$\mathbf{A}\mathbf{x} = \gamma\mathbf{B}\mathbf{x} \Leftrightarrow (\mathbf{A} - \sigma\mathbf{B})\mathbf{x} = (\gamma - \sigma)\mathbf{B}\mathbf{x}. \quad (12)$$

Thus,

$$\mathbf{C}\mathbf{x} = \theta\mathbf{x} \quad (13)$$

where

$$\mathbf{C} = (\mathbf{A} - \sigma\mathbf{B})^{-1}\mathbf{B}$$

$$\theta = \frac{1}{\gamma - \sigma}.$$

Suppose that $(\tilde{\theta}_i, \tilde{\mathbf{y}}_i)$, $i = 1, 2, \dots, k$ are the eigenpairs of the matrix $\mathbf{H}_k$

$$\mathbf{H}_k\tilde{\mathbf{y}}_i = \tilde{\theta}_i\tilde{\mathbf{y}}_i \quad (14)$$

We claim that the iteration has converged when

$$h_{k+1,k}|\mathbf{e}_k^*\tilde{\mathbf{y}}_i| \leq \text{tol} \quad (15)$$

where tol is a user prescribed tolerance.

In the application of the leaky-wave antennas, the searching range of the eigenvalues has to be enlarged because the leakage constants increase significantly in the working band. As a result, the Arnoldi method may converge very slowly, even not converge though the subspace is good enough. The refined shift-and-invert Arnoldi method [14]–[16] is introduced here to correct this problem. Since this method has a quite good convergence, a much simpler FDFD method with all six field components described before can be used though this FDFD method has a much large scale matrix.

D. Refined Shift-and-Invert Arnoldi Method

The refined method can be applied to the shift-and-invert Arnoldi method for the generalized eigenproblem. In order to speed the convergence rate, for each approximate eigenvalue available (Ritz value), seek a unit length refined Ritz vector from the subspace to approximate the eigenvector. The refined Ritz vector minimizes the corresponding residual norm formed with the approximate eigenvalue over the subspace [14]–[16]. For each $\tilde{\theta}_i$, seek a unit length vector $\mathbf{u}_i \in \mathbf{K}_k(\mathbf{C}, \mathbf{v}_1)$ that satisfies the following optimality property:

$$\|((\mathbf{A} - \sigma\mathbf{B})^{-1}\mathbf{B} - \tilde{\theta}_i\mathbf{I})\mathbf{u}_i\|$$

$$= \min_{\substack{\mathbf{u} \in \mathbf{K}_k(\mathbf{C}, \mathbf{v}_1) \\ \|\mathbf{u}\|=1}} \|((\mathbf{A} - \sigma\mathbf{B})^{-1}\mathbf{B} - \tilde{\theta}_i\mathbf{I})\mathbf{u}\|. \quad (16)$$

We use $(\tilde{\lambda}_i, \mathbf{u}_i)$ to approximate the eigenpair (λ_i, φ_i) of $(\mathbf{A}, \mathbf{B})$. Thus, $\mathbf{u}_i$ is the best approximation φ_i from $\mathbf{K}_k(\mathbf{C}, \mathbf{v}_1)$ with respect to $\tilde{\theta}_i$ and the Euclidean norm. The residual norm of the refined approximate eigenpair $(\tilde{\lambda}_i, \mathbf{u}_i)$ is given by [14], [15],

$$\|\hat{\mathbf{r}}_i\| = \|(\mathbf{A} - \tilde{\lambda}_i \mathbf{B})\mathbf{u}_i\|. \quad (17)$$

The refined Ritz vector $\mathbf{u}_i$ can be calculated by [15], [16]

$$\mathbf{u}_i = \mathbf{V}_k \mathbf{z}_i \quad (18)$$

where $\mathbf{z}_i$ is the right singular vector associated with the smallest singular value $\sigma_{\min}(\tilde{\mathbf{H}}_k - \tilde{\theta}_i \tilde{\mathbf{I}}_k)$ of the $(k+1) \times k$ Hessenberg matrix $\tilde{\mathbf{H}}_k - \tilde{\theta}_i \tilde{\mathbf{I}}_k$. It can be proved [15] that the residual norm satisfies

$$\|\hat{\mathbf{r}}_i\| \leq |\tilde{\lambda}_i - \sigma| \|(\mathbf{A} - \sigma \mathbf{B})\| \sigma_{\min}(\tilde{\mathbf{H}}_k - \tilde{\theta}_i \tilde{\mathbf{I}}_k). \quad (19)$$

As a result, we claim that the refined iteration has converged when

$$\sigma_{\min}(\tilde{\mathbf{H}}_k - \tilde{\theta}_i \tilde{\mathbf{I}}_k) \leq \text{tol} \quad (20)$$

where tol is a user prescribed tolerance and can be set as $10^{-6} - 10^{-8}$.

The refined iteration process is listed as follows:

- 1) Start with an arbitrary vector $\mathbf{v}_1$ with norm 1 and choose a shift σ , the dimension k of the Krylov subspace and a tolerance tol.
- 2) Repeat for $i = 2, 3, \dots$
 - Calculate $\mathbf{v}_i \leftarrow (\mathbf{A} - \sigma \mathbf{B})^{-1} \mathbf{B} \mathbf{v}_{i-1}$.
 - For i from 1 to $k-1$, $h_{i,k-1} \leftarrow \mathbf{v}_i^* \mathbf{v}_k$, $\mathbf{v}_k \leftarrow \mathbf{v}_k - h_{i,k-1} \mathbf{v}_i$.
 - $h_{k,k-1} \leftarrow \|\mathbf{v}_k\|$.
 - $\mathbf{v}_k \leftarrow (\mathbf{v}_k) / (h_{k,k-1})$.
 - Calculate the Ritz eigenpair $(\tilde{\theta}_i, \tilde{\mathbf{y}}_i)$ of the matrix $\mathbf{H}_k$.
- 3) Compute the approximate solutions $\tilde{\lambda}_i$ and $\mathbf{u}_i$, and use them to approximate the eigenpairs (λ_i, φ_i) of $(\mathbf{A}, \mathbf{B})$.
- 4) If all the residual norms $\|\hat{\mathbf{r}}_i\|$ are below tol, then stop, else continue.
- 5) Construct a new initial unit length vector $\mathbf{v}_1$ and go to Step 2.

III. SIMULATION AND EXPERIMENT

In order to validate the proposed method, at first an infinitely long rectangular waveguide periodically loaded with a series of lossy dielectric blocks as shown in Fig. 2 is analyzed. In the case of a dominant mode TE₁₀ in the waveguide, the complex propagation constants can be analytically calculated.

When the transfer matrix $\mathbf{T}$ of one periodic unit is obtained by using the Floquet's theorem [3], the complex propagation constant can be derived as

$$\gamma = \frac{1}{d} \cdot \cosh^{-1} \left(\frac{T_{11} + T_{22}}{2} \right). \quad (21)$$

In the presented example using the FDFD method to simulate the periodic guided-wave structure in Fig. 1, two group

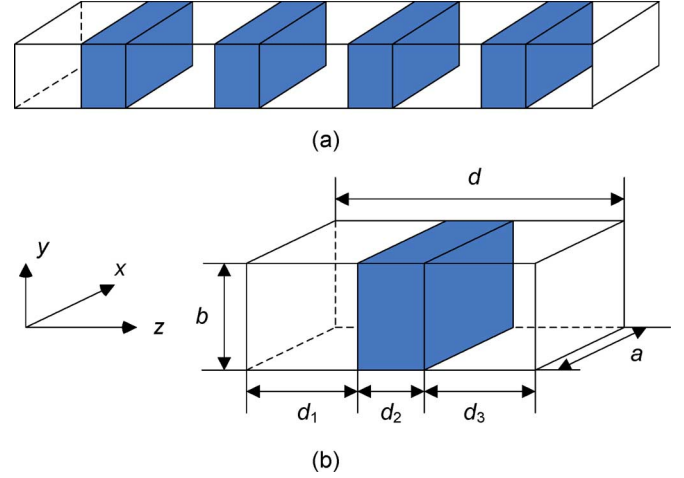


Fig. 2. An infinitely long rectangular waveguide loaded periodically by a series of lossy dielectric blocks. (a) Parts of infinite periodic guided-wave structure. (b) Single periodic cell structure model.

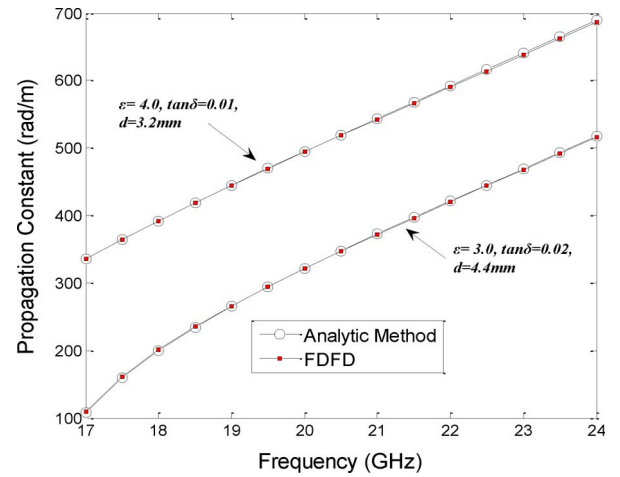


Fig. 3. Comparison of the propagation constants of TE₁₀ mode in the waveguide filled with lossy dielectric blocks between the FDFD method and the analytic method when $a = 6.8$ mm, $b = 1.2$ mm and $d_2 = 1.6$ mm.

dimensions are investigated. The first group dimensions of the waveguide are $a = 6.8$ mm, $b = 1.2$ mm, $d = 3.2$ mm, and $d_2 = 1.6$ mm. The dielectric constant $\epsilon_r = 4.0$ and a loss tangent $\tan \delta = 0.01$ is considered. The second group dimensions are $a = 6.8$ mm, $b = 1.2$ mm, $d = 4.4$ mm, and $d_2 = 1.6$ mm, $\epsilon_r = 3.0$, and $\tan \delta = 0.02$. The discretization is selected to $\Delta x = \Delta y = \Delta z = 0.4$ mm.

In Figs. 3 and 4, the results of both phase and attenuation constants of two group dimensions are shown respectively, which are calculated by using the proposed method and analytical method (21). Good agreements can be observed from these figures.

Next, the FDFD method is used in the application of periodic leaky-wave antennas. In Fig. 5, a periodic leaky-wave antenna based on a metal-strip-loaded dielectric inset waveguide is shown. This type of open periodic waveguide supports the propagation of leaky-wave modes, which can be used to set up leaky-wave antenna. Due to its desirable properties, the leaky-wave antenna is suitable to be used in millimeter waveband. This leaky-wave antenna has been analyzed by using the

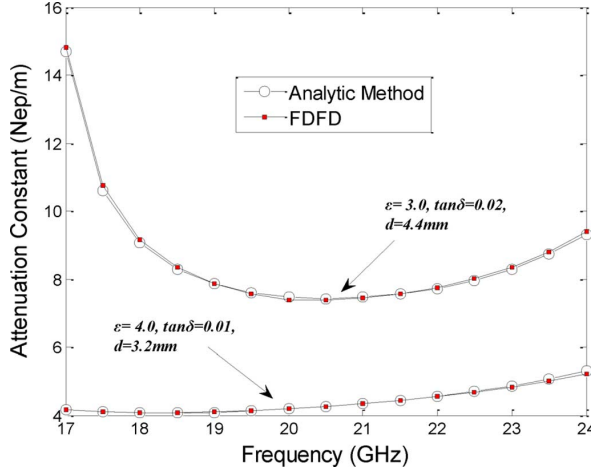


Fig. 4. Comparison of the attenuation constants of TE₁₀ mode in the waveguide filled with lossy dielectric blocks between the FDTD method and the analytic method when $a = 6.8$ mm, $b = 1.2$ mm and $d_2 = 1.6$ mm.

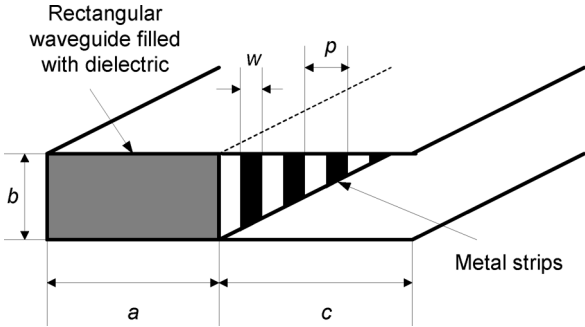


Fig. 5. Dielectric-inset waveguide leaky-wave antenna.

transverse equivalent network method [22] and full-wave spectral-domain method [23]. Obviously, the FDTD method can also be used to analyze this antenna based on the versatility of the FDTD method.

As shown in Fig. 5, the wide wall of rectangular waveguide is a and the narrow wall is b . One of the narrow walls is replaced by periodic metal strips. The width of metal strip is w and periodic length is p . The results of the lowest mode in [22] are used to compare with the results of the FDTD method. The sizes of one example ([22, Fig. 9]) are: $a = 1.4$ mm, $b = 1.4$ mm, $c/a = 2.0$, $(p - w)/p = 0.3$.

When the FDTD method combined with the refined shift-and-invert Arnoldi techniques is used to analyze this case of the leaky-wave antenna, the phenomena of the very slow convergence and even failure to convergent disappear. These phenomena will appear when only the normal shift-and-invert Arnoldi technique is used. The average calculation time of each simulation is about 70 seconds in a dual-core CPU of 1.66 GHz.

Fig. 6 shows the angle of maximum radiation when frequency changes, which can be calculated according to the equation of [24]

$$\sin \theta_m \approx \frac{\beta}{k_0} \quad (22)$$

where θ_m is the angle of maximum radiation, measured from the broadside direction. The backward radiation phenomena can

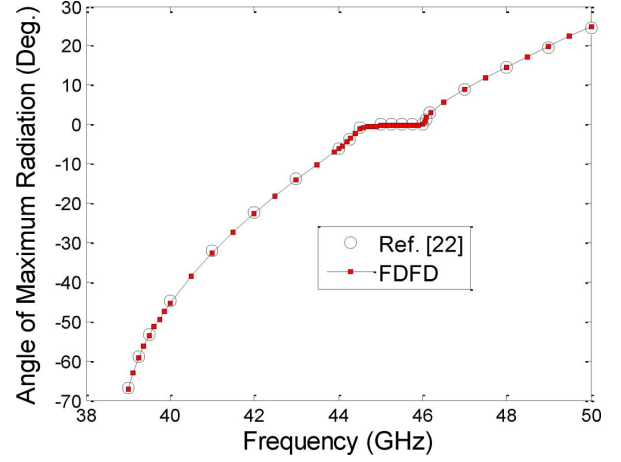


Fig. 6. Comparison of the angles of maximum radiation of the leaky-wave antenna.

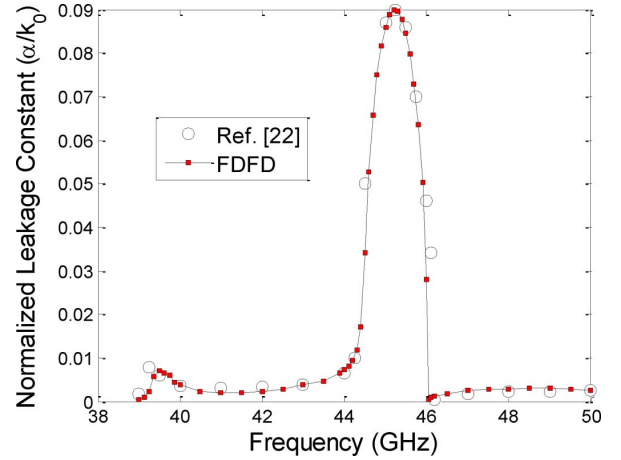


Fig. 7. Comparison of the normalized leakage constants of the leaky-wave antenna.

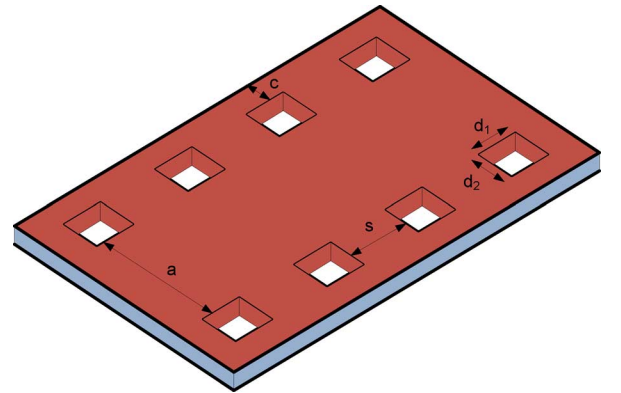


Fig. 8. Periodic leaky-wave antenna of SIW version.

be found from Fig. 6. Fig. 7 shows the comparison of normalized leakage constants. From the figures, the agreement between these results can be observed.

An example of the SIW version periodic leaky-wave antennas is presented below. As shown in Fig. 8, if the distance of the metalized via-holes increases, the SIW guided-wave structure will transform to a periodic leaky-wave antenna. The TE₂₀ mode has a better performance than TE₁₀ mode for this periodic leaky-wave antenna [25]. In Fig. 8, the dimensions are selected to an SIW width $a = 10$ mm, rectangular metal slot dimensions

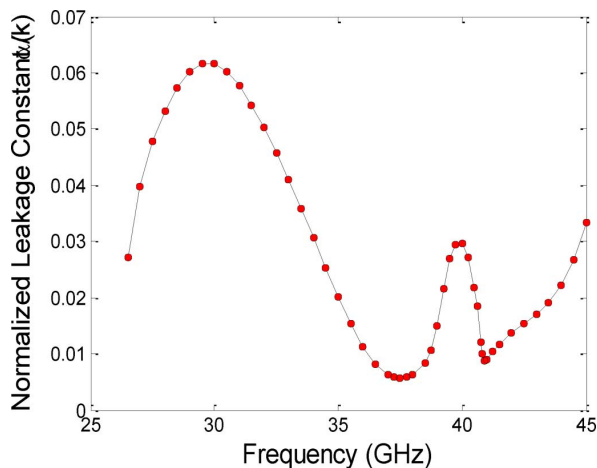


Fig. 9. Normalized leakage constant of the SIW leaky-wave antenna for the TE₂₀ mode, $s = 4.8$ mm, $d_1 = 0.8$ mm, and $c = 2.2$ mm.

$d_1 = 0.8$ mm and $d_2 = 0.8$ mm, and a distance between the slots $s = 4.8$ mm. This yields a period length $d = s + d_1 = 5.6$ mm. The distance between holes and the edge of the substrate c is 2.2 mm. An RT/duroid 5880 substrate with a thickness t of 0.787 mm, a permittivity $\epsilon_r = 2.2$, and a loss tangent $\tan \delta = 0.009$ is selected in the simulation.

Similarly, when using the FDFD method combined with the refined shift-and invert Arnoldi techniques to analyze this SIW leaky-wave antenna, the phenomena of the very slow convergence and some failures disappear. The simulation speed has been improved greatly. The Fig. 9 shows the normalized leakage constant of TE₂₀ mode and the central working frequency of the antenna is 31 GHz.

IV. CONCLUSION

A FDFD method combined with the refined shift-and invert Arnoldi technique is proposed to more accurately and fast analyze the characteristics of the periodic guided-wave structures and periodic leaky-wave antennas. By means of the refined shift-and invert Arnoldi technique, the improved FDFD method can directly solve a large scale generalized nonsymmetrical eigenvalues problem and speed the convergence of computing the desired eigenvalues and eigenvectors. Consequently, a simple difference FDFD method with the original six field components can be adopted. The numerical simulations have demonstrated that the improved method can also convergent very well in the working band of the periodic leaky-wave antennas. It means that when the attenuation constant or phase constant has a singular change, the improved FDFD method can also work. This makes the method be very useful when some periodic structures which use a singular guided-wave characteristic such as artificial magnetic conductors, metamaterials and so on, need to be analyzed.

REFERENCES

- [1] K. Wu, "Integration and interconnect techniques of planar and non-planar structures for microwave and millimeter-wave circuits-current status and future trend," in *Proc. Asia-Pacific Microw.*, Taipei, Taiwan, 2001, pp. 411–416.
- [2] F. Xu and K. Wu, "Guided-wave and leakage characteristics of substrate integrated waveguide," *IEEE Trans. Microw. Theory Tech.*, vol. 53, no. 1, pp. 66–73, Jan. 2005.
- [3] R. E. Collin, *Field Theory of Guided Waves*. New York, NY, USA: McGraw-Hill, 1960.
- [4] P. J. B. Claricoats and M. I. Sobhy, "Propagation behavior of periodically loaded waveguides," in *Proc. Inst. Elect. Eng.*, 1968, vol. 115, pp. 652–661.
- [5] W. Platte, "Microwave measurements of effective dielectric constant of semiconductor waveguides via periodic-structure photoexcitation," *IEEE Trans. Instrum. Meas.*, vol. 46, no. 3, pp. 717–721, Jun. 1997.
- [6] Md. R. Amin, K. Ogura, H. Kitamura, K. Minami, T. Wannabe, Y. Carmel, W. Main, J. Weaver, W. W. Destler, and V. L. Granatstein, "Analysis of the electromagnetic waves in an overmoded finite length slow wave structure," *IEEE Trans. Microw. Theory Tech.*, vol. 43, no. 4, pp. 815–822, Apr. 1995.
- [7] A. Zeid and H. Baudrand, "Electromagnetic scattering by metallic holes and its applications in microwave circuit design," *IEEE Trans. Microw. Theory Tech.*, vol. 50, no. 4, pp. 1198–1206, Apr. 2002.
- [8] F. Xu, Y. Zhang, W. Hong, K. Wu, and T. J. Cui, "Finite-difference frequency-domain algorithm for modeling guided-wave properties of substrate integrated waveguide," *IEEE Trans. Microw. Theory Tech.*, vol. 51, no. 11, pp. 2221–2227, Nov. 2003.
- [9] F. Xu, L. Li, K. Wu, S. Delprat, and M. Chake, "Parameter extraction of interdigital slow-wave coplanar waveguide circuits using finite difference frequency domain algorithm," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 18, no. 3, pp. 250–259, May 2008.
- [10] F. Xu, K. Wu, and W. Hong, "Equivalent resonant cavity model of periodic guided-wave structures and its application in finite difference frequency domain algorithm," *IEEE Trans. Microw. Theory Tech.*, vol. 55, no. 4, pp. 697–702, Apr. 2007.
- [11] Y. Saad, "Numerical methods for large eigenvalue problems," in *Algorithms and Architectures for Advanced Scientific Computing*. Manchester, U.K.: Manchester Univ. Press, 1992.
- [12] M. N. Kooper, H. A. van der Vorst, S. Poedts, and J. P. Goedbloed, "Application of the implicitly updated Arnoldi method with a complex shift-and-invert strategy in MHD," *J. Computat. Phys.*, vol. 118, pp. 320–332, May 1995.
- [13] R. B. Morgan, "On restarting the Arnoldi method for large nonsymmetric eigenvalue problems," *Math. Comput.*, vol. 65, no. 215, pp. 1213–1230, Jul. 1996.
- [14] Z. Jia, "A refined subspace iteration algorithm for large sparse eigenproblems," *Appl. Numer. Math.*, vol. 32, pp. 35–52, Nov. 1999.
- [15] Z. Jia, "Refined iterative algorithms based on Arnoldi's process for large unsymmetric eigenproblems," *Linear Algebra Appl.*, vol. 259, pp. 1–23, Jul. 1997.
- [16] Z. Jia and Y. Zhang, "A refined shift-and-invert Arnoldi algorithm for large unsymmetric generalized eigenproblems," *Comput. Math. With Applicat.*, vol. 44, pp. 1117–1127, Oct.–Nov. 2002.
- [17] W. E. Arnoldi, "The principle of minimized iterations in the solution of the matrix eigenvalues problem," *Quart. Appl. Math.*, vol. 9, pp. 17–29, 1951.
- [18] T. Ericsson and A. Rube, "The spectral transformation Lanczos method for the numerical solution of large sparse generalized symmetric eigenvalue problems," *Math. Comput.*, vol. 35, pp. 1251–1268, 1980.
- [19] R. G. Grimes, J. G. Lewis, and H. D. Simon, "A shifted block Lanczos algorithm for solving sparse symmetric generalized eigenproblems," *SIAM J. Matrix Anal. Appl.*, vol. 15, pp. 228–272, 1994.
- [20] D. C. Sorensen, "Implicit application of polynomial filters in a k-step Arnoldi method," *SIAM J. Matrix Anal. Appl.*, vol. 13, pp. 357–385, 1992.
- [21] R. B. Lehoucq and D. C. Sorensen, "Deflation techniques for an implicitly re-started Arnoldi iteration," *SIAM J. Matrix Anal. Appl.*, vol. 17, pp. 789–821, 1996.
- [22] M. Guglielmi and G. Boccia, "A novel theory for dielectric-inset waveguide leaky-wave antennas," *IEEE Trans. Antennas Propag.*, vol. 39, no. 4, pp. 497–504, Apr. 1991.
- [23] J. L. Gómez, F. D. Quesada, and A. A. Melcón, "Analysis and design of periodic leaky-wave antennas for millimeter waveband in hybrid waveguide-planar technology," *IEEE Trans. Antennas Propag.*, vol. 53, no. 9, pp. 2834–2842, Sep. 2005.
- [24] A. A. Oliner and R. C. Johnson, *Leaky-Wave Antennas, Antenna Engineering Handbook*, 3rd ed. New York, NY, USA: McGraw-Hill, 1993, ch. 10.
- [25] F. Xu, K. Wu, and X. Zhang, "Periodic leaky-wave antenna for millimeter wave applications based on substrate integrated waveguide," *IEEE Trans. Antennas Propag.*, vol. 58, no. 2, pp. 340–347, Feb. 2010.



Feng Xu (M'05–SM'09) was born in Jiangsu, China. He received the B.S. degree in radio engineering from Southeast University, Nanjing, China, in 1985, the M.S. degree in microwave and millimeter-wave theory and technology from the Nanjing Research Institute of Electronics and Technology, Nanjing, in 1998, and the Ph.D. degree in radio engineering from Southeast University, Nanjing, in 2002.

From 1985 to 1996, he was with the Nanjing Research Institute of Electronics and Technology, where he conducted research in the areas of antenna and RF circuits design. From 2002 to 2010, he was with the Poly-Grames Research Center, École Polytechnique, Montréal, QC, Canada, where he was a Post-Doctoral Researcher and was then a Research Associate. In April of 2010, he joined the R&D Department of the Safe Engineering Services and Technologies, Montréal, QC, Canada, where he was Senior Researcher Scientist of the R&D Department. Since January of 2011, he has been a Professor at the College of Electronic Science and Engineering, Nanjing University of Posts and Telecommunication, sponsored by the Specially Appointed Professor Program of Jiangsu Province of China. His current research interests include numerical methods, the modeling and design of advanced microwave and millimeter-wave circuits and components, and microwave photonic circuits and systems. He is also interested in transient electromagnetic scattering, electromagnetic interference (EMI) and electromagnetic compatibility (EMC).



Wei Hong (M'92–SM'07–F'12) received the B.S. degree from the University of Information Engineering, Zhengzhou, China, in 1982, and the M.S. and Ph.D. degrees from Southeast University, Nanjing, China, in 1985 and 1988, respectively, all in radio engineering.

Since 1988, he has been with the State Key Laboratory of Millimeter Waves and serves for the Director of the lab since 2003, and is currently a Professor and the vice-dean of the School of Information Science and Engineering, Southeast University. In 1993, 1995, 1996, 1997, and 1998, he was a short-term Visiting Scholar with the University of California at Berkeley and at Santa Cruz, respectively. He has been engaged in numerical methods for electromagnetic problems, millimeter-wave theory and technology, antennas, electromagnetic scattering, RF technology for wireless communications, etc. He has authored and coauthored over 200 technical publications, and authored two books of *Principle and Application of the Method of Lines* (in Chinese, Southeast Univ. Press, 1993) and *Domain Decomposition Methods for Electromagnetic Problems* (in Chinese, Science Press, 2005).

Dr. Hong was thrice awarded the first-class Science and Technology Progress Prizes issued by the Ministry of Education of China and Jiangsu Province Government, etc. Besides, he also received the Foundations for China Distinguished Young Investigators and for "Innovation Group" issued by NSF of China. He is a Vice-President of the Microwave Society and Antenna Society of CIE, Chairperson of IEEE MTT-S/AP-S/EMC-S Joint Nanjing Chapter, and served as an Associate Editor of the IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUE during 2007–2010, the Associate-Editor-in-Chief for *China Communications*, and an editor board member for *IJAP*, and the *Chinese Science Bulletin*, etc.